

Spectral Clustering:

- Spectral clustering is a powerful and versatile clustering technique that uses concepts from graph theory and linear algebra to detect groups (clusters) in complex data.
- Unlike traditional methods such as k-means, spectral clustering does not rely solely on geometric distances or convex shapes.
- Highly effective for identifying non-linear and arbitrarily shaped clusters.

Procedure:

- The process begins by representing the data as a similarity graph, where each node is a data point and edges signify similarity between pairs of points.
- From this graph, a matrix called the Laplacian is constructed, which encodes both the connectivity and affinity between nodes.
- The core idea of spectral clustering is to compute the eigenvalues and eigenvectors (also called spectrum) of this Laplacian matrix.
- By projecting the data onto the space formed by a subset of its key eigenvectors, spectral clustering transforms the original problem into one that is easier to solve with algorithms like k-means.

Simple Example of Laplacian in Spectral Clustering

Suppose we have 4 data points that form two clear groups:

- **Points:** A, B, C, D
- **Similarities:** A is strongly connected to B; C is strongly connected to D; there are no strong connections between (A,B) and (C,D).

We can represent these connections as a graph:

- A -- B C -- D

Adjacency Matrix (W):

$$W = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

Degree Matrix (D): (each diagonal entry is the sum of the connections for that node)

$$D = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Laplacian Matrix (L):

$$L = D - W = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

Key Step:

Spectral clustering computes the eigenvectors of this Laplacian. In this example, the first non-trivial (non-constant) eigenvector separates the points into two clusters: (A, B) and (C, D). If you plot these points using this eigenvector, A and B group together, as do C and D—revealing the two clusters.

Examples:

image segmentation, social network analysis, and bioinformatics.

Graph clustering is a technique used to identify groups of densely connected nodes (called clusters or communities) within a graph. In real-world networks—such as social, biological, transportation, or citation networks—nodes represent entities, and edges represent relationships among them.

The main objective of graph clustering is to partition the graph so that nodes within the same group have many connections among themselves and few connections to nodes outside their group.

To perform graph clustering, various algorithms are used, such as Markov Clustering (MCL), Spectral Clustering,

These methods consider the connectivity patterns of nodes and may utilize mathematical tools, such as random walks, to identify clusters.

1. Random Walks / Markov Clustering

Consider a graph with five nodes:

- Nodes: 1, 2, 3 are tightly connected (edges: 1-2, 2-3, 1-3).
- Nodes: 4 and 5 are only connected (edge: 4-5).

If you start a random walk at node 1, you are very likely to keep visiting 2 or 3, but rarely 4 or 5.

Markov Clustering simulates this flow and separates the graph into two clusters: {1,2,3} and {4,5}.

Graph clustering is significant in many fields:

- **Social networks:** Detect communities of closely associated users.
- **Biological networks:** Identify functional modules, such as protein complexes.
- **Information retrieval:** Group similar webpages or documents.